

Week 9 A Fear and Greed Index from News Headlines

FINS5545: FINANCIAL MARKET DATA LITERACY

TODAY'S LECTURE

We turn the project news headlines into a **fear and greed index**, a number that measures how fearful or greedy the market is.

- We compute a sentiment score for each headline with two models, **VADER** from Week 8 and **finVADER**, a version of VADER extended with finance word lists.
- We rescale that score onto a 0 to 100 scale, where 0 is extreme fear and 100 is extreme greed.
- We average it across all 50 stocks, within each industry, and for a single stock.
- We then standardise the index, which is what makes it possible to say that a given day was unusually fearful/greedy.

A fear and greed index is the average sentiment score of many headlines on a 0 to 100 scale, and it allows us to gauge market sentiment.

SETUP BEFORE WE START

Download the Week 9 scripts from Ed and unzip them into your fins2026 folder.

```
fins2026/  
  week9/  
    fear_greed_index/  
      01_recap_vader_meet_finvader.py ... 05_make_fear_greed_figures.py  
      fear_greed_tools.py  
      news_headlines.parquet           the project headlines  
      output/                          created when you run the scripts
```

This is the same headline file we used in Week 7 – nothing new

PACKAGES USED TODAY

If you cannot run the code today, install the packages below, or ask your AI coding agent to install them.

```
pip install vaderSentiment finvader pandas matplotlib pyarrow
```

Package	Job today
vaderSentiment	the general-purpose VADER model
finvader	VADER plus two finance word lists
pandas, pyarrow	tables and the headline file
matplotlib	the figures

The `finvader` package depends on `nltk`, which `pip` installs alongside it, and the VADER word list downloads the first time you run it. You should already have this installed.

GLOSSARY (1) — FEAR, GREED, AND SENTIMENT

Term	Plain meaning
sentiment	the tone of a piece of writing, how positive or negative it is
compound score	VADER's summary sentiment number for a text, from -1 to $+1$
fear	the news is negative in tone, so the index falls below 50
greed	the news is positive in tone, so the index rises above 50
fear and greed index	the average sentiment score of many headlines
standardised index	the index minus its mean, divided by its standard deviation, so 0 is a normal day
contrarian	acting against the crowd, buying at extreme fear and selling at extreme greed

GLOSSARY (2) — BUILDING THE INDEX

These are the parts we put together to build the index.

Term	Plain meaning
VADER	Valence Aware Dictionary and sEntiment Reasoner, the Week 8 model
finVADER lexicon	VADER with two finance word lists added a fixed list of words. In a sentiment lexicon every word carries a numeric valence, its rated positivity or negativity
aggregate per industry	an index across all 50 stocks, pooling every headline that day an index for each of the 10 sectors, using that sector's headlines
per stock coverage	an index for a single stock, using only that stock's headlines the share of days on which a group actually has news
rolling average	the mean over a moving window of days, used to smooth a noisy line

WHAT A FEAR AND GREED INDEX IS

INVESTOR SENTIMENT AND ASSET PRICES

Classical asset pricing assumes that price equals fundamental value, because rational investors trade away any gap. The sentiment literature relaxes that assumption.

- Baker and Wurgler (2007) define **investor sentiment** as a belief about future cash flows and investment risk that the facts available at the time do not justify.
- Arbitrage against such a belief is costly and risky, so a mispricing can persist instead of being corrected straight away.
- In their evidence sentiment moves prices most for stocks that are hard to value and hard to arbitrage, such as small, young, and unprofitable firms.

Sentiment moves prices, and it is measurable from text, which is why constructing a quantitative measure of it is worthwhile.

THE CNN FEAR AND GREED INDEX

The best-known example is the **fear and greed index** published by CNN Business (2026).

- It spans the range 0 to 100. Below 25 is extreme fear, near 50 is neutral, above 75 is extreme greed.
- It combines seven market signals, such as price momentum, demand for safe assets, and market volatility.
- Each of the seven is measured against its own recent history before being combined, a step we will need ourselves later.
- Many investors treat it as a **contrarian** signal, reading extreme fear as a possible bottom and extreme greed as a possible top.

The CNN index is computed from market data such as prices, flows, and volatility. The index we build is computed from the text of news headlines.

A SENTIMENT INDEX FOR STOCKS

The same idea can be constructed directly from text, by computing a sentiment score for each article and then averaging those scores.

- Tetlock (2007) finds that unusually pessimistic news predicts falling prices, followed by a partial reversion.
- The San Francisco Federal Reserve publishes a Daily News Sentiment Index built this way (Shapiro et al., 2022).
- The construction is the same in every case. Compute a sentiment score for every article, then average those scores by day, by sector, or across all stocks.

USES OF A NEWS SENTIMENT INDEX

A news sentiment index is used because it is available daily and in real time, while survey measures of sentiment are monthly and arrive with a lag.

- Shapiro et al. (2022) find that news sentiment in the days before a consumer confidence release strongly predicts that release.
- In the same study a positive sentiment shock raises consumption and output, with the effect building over the following 12 months.
- Tetlock (2007) finds that high media pessimism predicts downward pressure on market prices, which then reverts.
- Extreme sentiment can persist for long periods, so the index informs a decision rather than dictating a trade.

News sentiment has measurable predictive power for survey sentiment, consumption, output, and short-horizon market returns.

WHAT WE BUILD THIS WEEK

We build our own fear and greed index from the project headlines, the same file from Week 7.

Item	This week's data
Headlines	149,683
Stocks	50
Industries	10
Window	2020 to 2023

We compute a sentiment score for every headline, rescale it to a 0 to 100 scale, average it three ways, and finally standardise the result. This is Stage 3 of the Data Factory Floor, turning cleaned text into a decision-ready number.

VADER AND FINVADER

VADER is the model we built in Week 8 (Hutto and Gilbert, 2014).

- It uses a lexicon of about 7,500 everyday words, each carrying a human-rated valence from -4 to $+4$.
- It applies five rules for capital letters, exclamation marks, boosters, negation, and the word **but**.
- It returns a **compound score** from -1 to $+1$, the number we use today.

VADER turns any text into a compound score from -1 to $+1$ using a fixed word list and five rules.

WHERE VADER STRUGGLES ON FINANCE TEXT

VADER was built for social media text rather than company news, so finance vocabulary is poorly covered.

- Many finance words are absent from its lexicon, so `guidance`, `hawkish`, and `impairment` contribute nothing to the compound score.
- Loughran and McDonald (2011) showed that most words classified as negative by a general dictionary, such as `liability` and `tax`, are not negative in a finance context.
- A compound score of zero can therefore mean the text is genuinely neutral, or that none of its words appear in the lexicon.

On finance text a general word list leaves gaps, so we add a finance word list.

WHAT FINVADER ADDS

finVADER is the same VADER model with two finance word lists added (Koráb, 2023).

Word list	Terms	What it covers
SentiBigNomics	7,295	economics and finance terms (Consoli et al., 2022)
Henry's list	189	words from earnings releases (Henry, 2008)

finVADER keeps VADER's five rules and its everyday words, then adds these finance words on top. Because its lexicon covers finance vocabulary, it is the model we build the index with.

finVADER is VADER plus about 7,500 finance words, so it covers the vocabulary of company news far more completely.

INSTALLING AND USING finVADER

finVADER installs with `pip` and needs Python and the `nltk` package.

```
from finvader import finvader

text = "Nvidia beats earnings, shares surge to a record high"
score = finvader(text,
                  use_sentibignomics=True,      # the 7,295-term list
                  use_henry=True,               # the 189-word list
                  indicator="compound")         # the -1 to 1 score
print(score)    # 0.8290
```

The compound indicator is the same -1 to $+1$ score VADER returns, so everything from Week 8 carries straight over.

ASK YOUR AI AGENT FOR FINVADER

If you get stuck, a clear request to your AI coding agent works well. State the goal, the input, and the output you want.

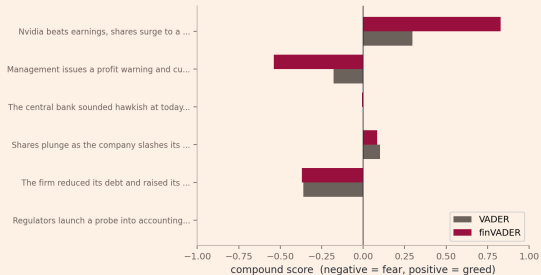
```
Install the finvader package. I have a pandas column of news headlines called title. Add a new column finvader_compound that holds each headline's finVADER compound score, using both finance word lists. Build the analyser once so it is fast on 100,000 rows, and show me the code.
```

Naming the column, both word lists, and the speed requirement gives the agent enough to return code you can read and run.

VADER AND finVADER ON SIX HEADLINES

The finance word list moves finance headlines

VADER and finVADER compound score, same six headlines



Source: worked example. VADER and finVADER compound score.

finVADER lifts `Nvidia beats earnings` from `+0.30` to `+0.83` and deepens the profit warning headline to `-0.54`. Both models get two of the six wrong. `Shares plunge` returns a slightly positive value, and `reduced its debt and raised its outlook` returns `-0.37`, because `reduced` and `debt` are negative words in isolation.

FROM A SCORE TO A 0 TO 100 SCALE

THE COMPOUND SCORE SPANS MINUS ONE TO ONE

Both models return a compound score over the same range.

- -1 is the most negative value a text can take, and $+1$ is the most positive.
- 0 means the positive and negative valences cancel, either because the text is neutral or because its words are absent from the lexicon.
- In Week 8 we stopped here. This week we rescale the number onto the fear and greed scale.

The compound score is signed, from -1 to $+1$, with 0 in the middle.

RESCALING THE SCORE TO ZERO TO ONE HUNDRED

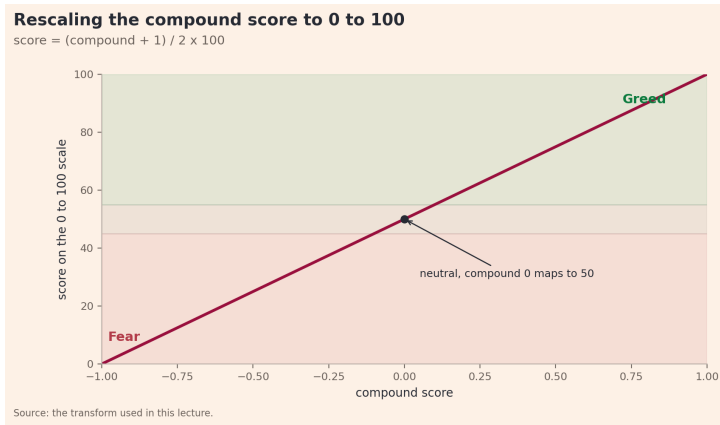
A fear and greed index is easier to interpret on the 0 to 100 scale, like the CNN index. We shift and stretch the compound score to get there.

$$\text{score} = \frac{\text{compound} + 1}{2} \times 100$$

compound	score	band
-1.0	0	extreme fear
-0.5	25	fear
0.0	50	neutral
+0.5	75	greed
+1.0	100	extreme greed

The transform is a straight line. A compound of 0 maps to 50, the neutral point.

THE SAME TRANSFORM AS A FIGURE



The 0 to 100 value is the compound score expressed on a different scale. The tone being measured is unchanged, only the units.

MOST INDIVIDUAL HEADLINES SIT NEAR FIFTY

Computing the finVADER score for all 149,683 headlines, most land close to neutral.

Band	Headlines	Share
Extreme fear (0 to 25)	2,447	1.6%
Fear (25 to 45)	14,180	9.5%
Neutral (45 to 55)	94,009	62.8%
Greed (55 to 75)	31,304	20.9%
Extreme greed (75 to 100)	7,743	5.2%

Nearly two-thirds of individual headlines are neutral, so the fear and greed signal only appears once we average many of them.

BUILDING THE INDEX

FROM HEADLINE SCORES TO AN INDEX

The index for a day is the mean 0 to 100 score over a set of headlines. The three levels below differ only in which headlines that set contains.

- **Aggregate**, every headline that day, across all 50 stocks.
- **Per industry**, the headlines in each of the 10 sectors that day.
- **Per stock**, the headlines for a single stock that day.

The calculation is the same mean in all three cases. Selecting a different set of headlines is what produces a market, an industry, or a stock index.

COVERAGE AND THE VOLATILITY OF THE INDEX

The daily mean is computed from however many headlines exist that day. When that count is small the mean moves a long way from one day to the next, because a few headlines carry it.

Level	Days with news	Share of days	SD of daily change
Single stock (median)	924	63%	10.02
Industry (median)	1,374	94%	8.32
Aggregate	1,461	100%	3.15

Out of 1,461 days in 2020 to 2023, a typical stock has news on about two days in three, an industry on almost every day, and all 50 stocks together on every day.

Averaging more headlines cuts the day-to-day movement of the index from 10.0 points to 3.2, which is why the aggregate and industry indices are the usable ones.

COMPUTING THE SENTIMENT SCORES IN PYTHON

We build the finVADER analyser once, compute a compound score for every headline, then rescale it to 0 to 100.

```
from fear_greed_tools import build_finvader, load_project_headlines,
    to_score_100

headlines = load_project_headlines()           # 149,683 rows
finvader = build_finvader()                   # built once, so it is fast

headlines["compound"] = [finvader.polarity_scores(t)["compound"]
                        for t in headlines["title"]]
headlines["score_100"] = to_score_100(headlines["compound"])
```

In the run-along we compute the distinct headline scores once and cache them, so all 149,683 rows are processed in a few seconds rather than a few minutes.

BUILDING THE DAILY INDEX IN PYTHON

Each level is a `groupby` followed by a `mean`.

```
# aggregate, every headline that day
aggregate = headlines.groupby("date")["score_100"].mean()

# per industry, split by the sector column
by_sector = headlines.groupby(["date", "sector"])["score_100"].mean()

# per stock, keep a single stock then group by day
nvda = (headlines[headlines["ticker"] == "NVDA"]
        .groupby("date")["score_100"].mean())
```

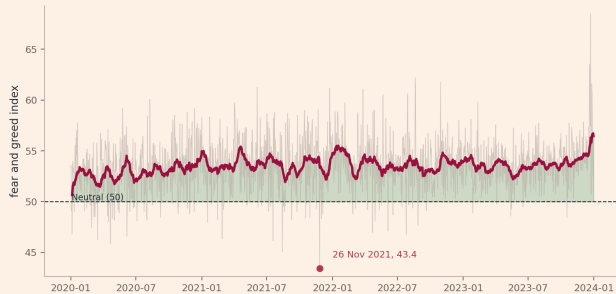
That is the index calculation. A 21-day rolling average then smooths the daily line so the trend is visible.

THE INDEX OUTPUT

THE AGGREGATE FEAR AND GREED INDEX

In levels the aggregate index is greedy on almost every day

finVADER index, all 50 stocks, daily (thin) and 21-day average (bold)



Source: project news headlines, 2020-2023. finVADER compound score.

Averaged over all 50 stocks the index stays close to 50, with a mean of **53.5**. Its lowest reading, 43.4, falls on 26 November 2021, the day the World Health Organization designated the Omicron COVID variant a variant of concern. The S&P 500 fell 2.3% that day, its largest one-day drop since the previous February.

IN LEVELS THE INDEX IS ALMOST ALWAYS GREEDY

Read against the CNN bands, the index we have built so far has a problem.

- It falls below 50 on only **6.3%** of the 1,461 days, because most headlines are mildly positive rather than balanced around zero.
- It therefore reports greed for four straight years, and the extreme fear band below 25 never once fires.
- A measure that returns the same answer every day carries no information about any particular day.

The level of the index tells us the news is mildly positive on average, which we already knew. It does not tell us which days were relatively more fearful or greedy.

STANDARDISE THE INDEX AGAINST ITS OWN HISTORY

The question worth asking is not whether the news is positive, but whether it is positive relative to normal for this index. We subtract the mean and divide by the standard deviation.

$$z_t = \frac{\text{index}_t - \overline{\text{index}}}{\text{sd}(\text{index})}$$

Band	Days	Share
Extreme fear ($z < -1.5$)	91	6.2%
Fear (-1.5 to -0.5)	315	21.6%
Neutral (-0.5 to $+0.5$)	657	45.0%
Greed ($+0.5$ to $+1.5$)	306	20.9%
Extreme greed ($z > +1.5$)	92	6.3%

Standardising spreads the days across all five bands, so the measure can now separate one day from another.

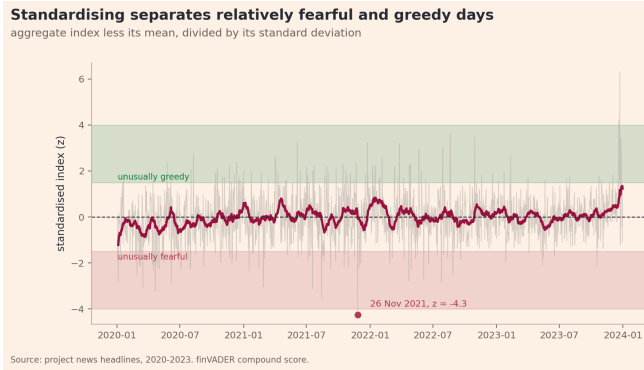
STANDARDISING IN PYTHON

Two lines, applied to the daily aggregate index.

```
mean, sd = aggregate.mean(), aggregate.std()  
z = (aggregate - mean) / sd
```

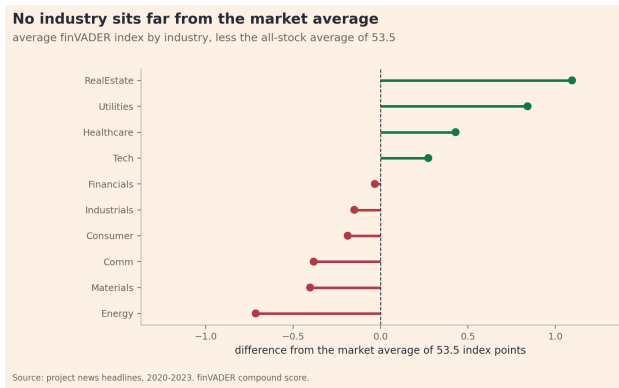
The mean and standard deviation here use the full sample, including days that had not happened yet. That is acceptable when describing history, which is all we do today. A live index would have to standardise against history known at the time, using an expanding window. For this series the two versions correlate at **0.996**, so the choice makes little difference here, but on a trading signal it would matter.

THE STANDARDISED INDEX



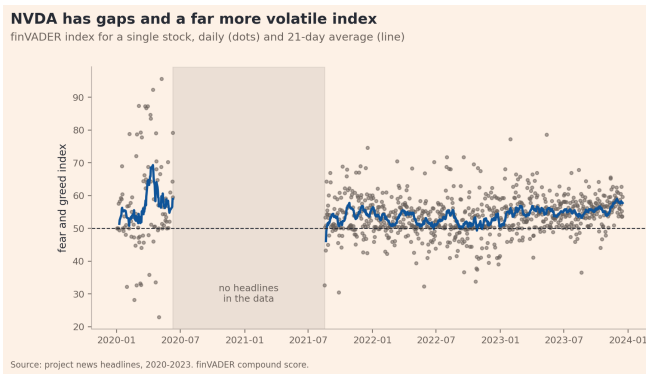
26 November 2021 is now a $z = -4.3$ reading, over four standard deviations below normal, rather than a number slightly below 50. CNN standardise each of their seven components in the same way before combining them, which is why their published index uses the full 0 to 100 range and our raw average does not.

FEAR AND GREED BY INDUSTRY



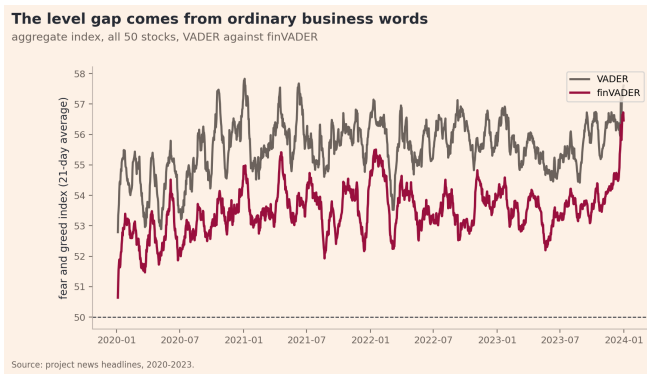
Measured against 50 every industry looks greedy, which tells us nothing, so the figure measures each against the market average of 53.5 instead. Real Estate is the most greedy at **+1.1** and Energy the least at **-0.7**. The full spread is 1.9 index points, against a standard deviation across individual headlines of 12.1, so industry is a small influence on the tone of a headline.

NVDA, A SINGLE-STOCK INDEX



The sentiment index is noisier for an individual stock. NVDA has no headlines in the data for a 433-day stretch in 2020 and 2021, then near-daily news after that, and its index moves three times as much from day to day as the aggregate.

VADER AGAINST finVADER



VADER is about **two points** higher throughout, 55.6 against 53.5. The gap is not noise. It traces to a small set of common words.

WHAT THE INDEX CAN AND CANNOT TELL YOU

It can	It cannot
average many headlines into a readable number	judge whether a headline is true or important
identify relatively more fearful or greedy days, once standardised	always get the sign right, as with the shares plunge headline
line up with known fear events, such as the Omicron drop	serve as a buy or sell rule on its own

- A fear and greed index is the average sentiment score of many headlines on a 0 to 100 scale.
- finVADER is VADER plus about 7,500 finance words, and its lexicon covers the vocabulary of company news.
- The 0 to 100 value is $(\text{compound} + 1)/2 \times 100$, so a compound of 0 maps to a neutral 50.
- Averaging more headlines cuts the index's day-to-day movement from 10.0 points to 3.2.
- In levels the index is above 50 on 94% of days, so it must be standardised before it can identify a relatively more fearful or greedy day.
- VADER and finVADER differ by two index points, because they disagree about whether words like `profit` and `energy` carry tone.

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